

Self-supervised Learning for AC Optimal Power Flow in Distribution Systems with Hard Constraints

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Overview

We address AC optimal power flow (AC-OPF) in distribution networks: compute controllable DER setpoints that minimize operational cost while satisfying **hard power-flow and equipment limits**.

We introduce two self-supervised learning pipelines and a fast Sequential Linearized Feasibility-Seeking (SLFS) module that projects predictions to feasible operating points.

Our approach achieves IPOPT-level feasibility with small optimality gaps and delivers **2.6–384×** runtime speedups on 33–1000-bus feeders.

Contributions

- **SLFS for hard constraints:** A Sequential Linearized Feasibility-Seeking procedure that enforces operational constraints via small convex subproblems.
- **Two self-supervised pipelines:**
 - **SLFS-FSNet** — incorporates SLFS during training and inference for strong feasibility;
 - **TPTF** — trains with fast penalty losses and applies SLFS at test time only when needed.
- **Theoretical guarantees:** Local convergence and monotone reduction of constraint violation under mild regularity assumptions.
- **Empirical validation:** IPOPT-level feasibility with small optimality gaps, **2.6–384×** speedups on 33/123/300/1000-bus feeders, and robustness to distribution shift.

Nonconvex AC-OPF Formulation

$$\begin{aligned} \min_{\mathbf{x}} f(\mathbf{x}) \\ \text{s.t. } \mathbf{l} \leq \mathbf{g}(\mathbf{x}) \leq \mathbf{u} & \quad (\text{operational constraints}) \\ \mathbf{x}^\top \mathbf{Q}_j \mathbf{x} \leq (s_j^{\text{der,max}})^2 \quad \forall j \in \mathcal{D} & \quad (\text{DER apparent-power limits}) \\ \mathbf{l}_x \leq \mathbf{x} \leq \mathbf{u}_x & \quad (\text{variable bounds}) \end{aligned}$$

Here $\mathbf{x}$ collects network state and controllable DER setpoints (e.g., bus voltages and DER real/reactive injections); $\mathbf{g}(\mathbf{x})$ includes voltage magnitudes/angles, line currents, and substation power; $\mathbf{l}, \mathbf{u}$ are constraint bounds; $\mathbf{Q}_j$ is a diagonal matrix encoding the apparent power at DER j and $s_j^{\text{der,max}}$ is its rated capacity.

Neural Network Design

Neural network architecture

Simple 4-layer fully-connected network with LeakyReLU activations.

Loss functions

- SLFS-FSNet: For a batch $\mathcal{S}$ and raw NN predictions $\mathbf{x}^{\text{raw}}$,

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} f(\mathbf{x}^{(j)}) + \gamma \left\| \mathbf{x}^{(j)} - (\mathbf{x}^{\text{raw}})^{(j)} \right\|_2^2$$

- TPTF: Using a penalty loss to mitigate constraint violation during training.

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} f(\mathbf{x}^{(j)}) + \rho c(\mathbf{x}^{(j)}),$$

$$c(\mathbf{x}^{(j)}) = \left\| [\mathbf{g}(\mathbf{x}^{(j)}) - \mathbf{u}]_+ + [\mathbf{l} - \mathbf{g}(\mathbf{x}^{(j)})]_+ \right\|_2^2$$

where the penalty weight ρ can be adaptively updated based on the degree of constraint violation.

Proposed Frameworks

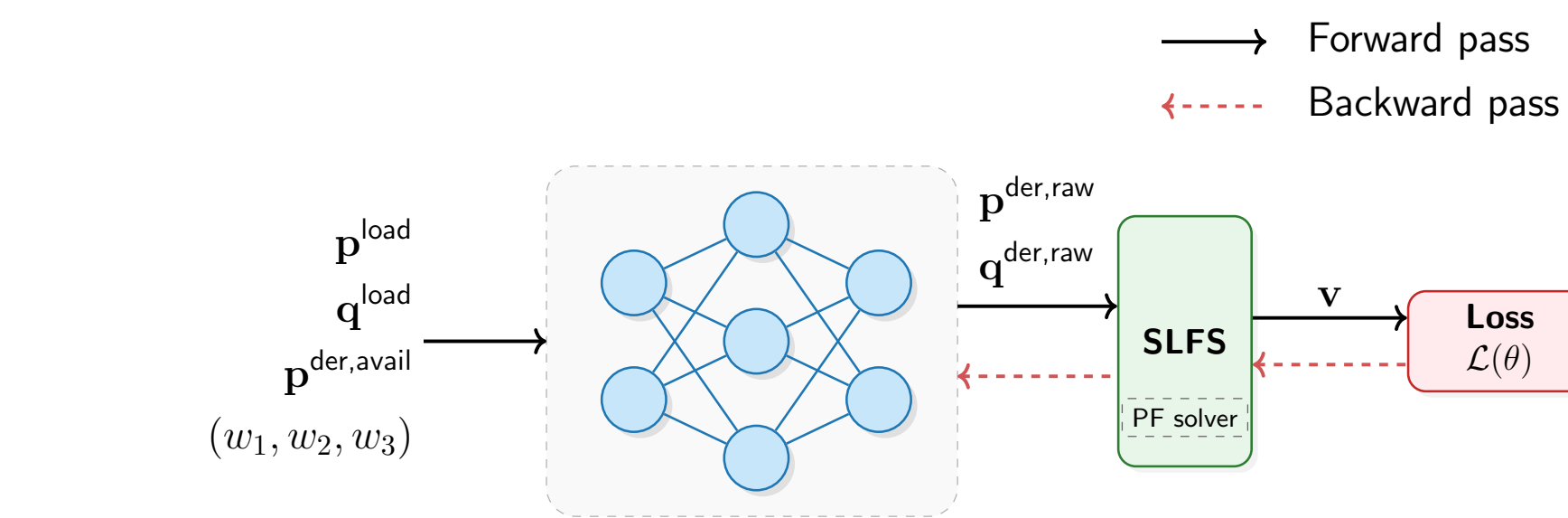


Figure 1. Feasibility-Seeking Neural Network (FSNet) framework. The SLFS module is used in both training and test time.

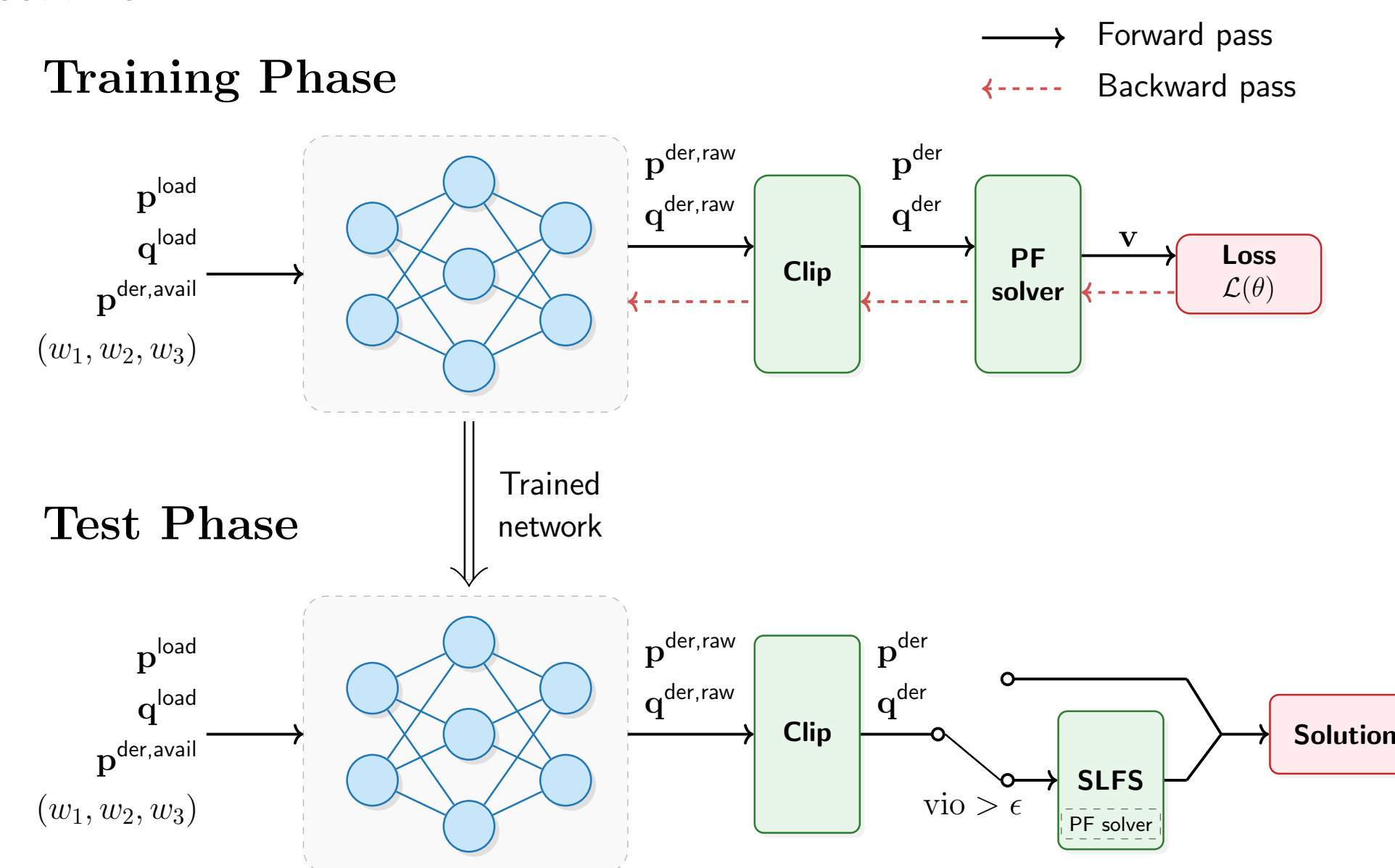


Figure 2. Train-Penalty and Test-Feasibility (TPTF) framework. The SLFS module is only used at test time if the constraint violation exceeds a certain threshold.

Sequential Linearized Feasibility-Seeking (SLFS)

Input: Predictions $\mathbf{p}^{\text{der}}, \mathbf{q}^{\text{der}}$ (possibly infeasible DER setpoints)

Output: Feasible DER setpoints $\mathbf{p}^{\text{der}*}, \mathbf{q}^{\text{der}*}$ satisfying all AC-OPF constraints

1. *Power flow solver:* Construct $\mathbf{x}_k$ from $\mathbf{p}^{\text{der}}, \mathbf{q}^{\text{der}}$ and solve power flow equations to get the corresponding network state (e.g., bus voltages, line currents).

$$\mathbf{v}_{k+1} = \mathbf{G}(\mathbf{v}_k, \mathbf{s})$$

2. *Linearize constraints:* Build a local linear model of constraint $\mathbf{g}(\mathbf{x}) \approx \mathbf{A}(\mathbf{x}_k)\mathbf{x} + \mathbf{b}(\mathbf{x}_k)$.
3. *Feasibility step:* Solve a small convex feasibility subproblem with the linear model.

$$\tilde{\mathbf{x}}_{k+1} = \underset{\mathbf{x}}{\text{argmin}} V^{\text{lin}}(\mathbf{x}; \mathbf{x}_k) + \frac{\eta}{2} \|\mathbf{x} - \mathbf{x}_k\|_2^2,$$

$$\text{s.t. } \mathbf{x}^\top \mathbf{Q}_j \mathbf{x} \leq (s_j^{\text{der,max}})^2, \forall j \in \mathcal{D}$$

$$\mathbf{l}_x \leq \mathbf{x} \leq \mathbf{u}_x$$

where $V^{\text{lin}}(\mathbf{x}; \mathbf{x}_k) = \left\| [\mathbf{A}(\mathbf{x}_k)\mathbf{x} + \mathbf{b}(\mathbf{x}_k) - \mathbf{u}]_+ + [\mathbf{l} - \mathbf{A}(\mathbf{x}_k)\mathbf{x} - \mathbf{b}(\mathbf{x}_k)]_+ \right\|_\infty$.

4. *Damped update:* Update the solution with a damped step $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha(\tilde{\mathbf{x}}_{k+1} - \mathbf{x}_k)$, where $\alpha \in (0, 1]$ is a step size.
5. Repeat steps 1–3 for a few iterations.

Main Theoretical Guarantee

Let the true violation function be $V(\mathbf{x}) = \left\| [\mathbf{g}(\mathbf{x}) - \mathbf{u}]_+ + [\mathbf{l} - \mathbf{g}(\mathbf{x})]_+ \right\|_\infty$.

Theorem. Under mild regularity assumptions and proper choice of α_k , the SLFS procedure guarantees monotone reduction of constraint violation at each iteration.

$$V(\mathbf{x}_{k+1}) \leq V(\mathbf{x}_k) - \gamma_k \|\tilde{\mathbf{x}}_{k+1} - \mathbf{x}_k\|^2$$

where $\gamma_k > 0$ depends on the local curvature of the constraints and the step size α_k .

Experimental Results

Takeaways

- Can achieve the similar degree of constraint satisfaction as IPOPT with small optimality gaps
- **2.6–384×** faster (larger grid = larger speedup)
- Training for TPTF is **3.2–11.6×** faster than SLFS-FSNet
- Our methods are **robust to distribution shift** between training and test data.

Experimental setup

- Datasets: 33/123/300/1000-bus feeders with 50k samples (including both infeasible and feasible instances) and 2k-4k test samples
- Baselines: Direct Penalty (no PF solver, no SLFS), Penalty (train with PF solver, no SLFS)
- Metrics: Mean/max constraint violation, optimality gap, runtime speedup over IPOPT

Detailed results

Training time (seconds)				
Method	33 Bus	123 Bus	300 Bus	1000 Bus
Direct Penalty	34.3	29.6	31.5	64.4
Penalty	32.6	30.3	29.6	62.2
TPTF	37.4	31.3	27.2	55.3
SLFS-FSNet	121.2	129.7	173.1	638.6

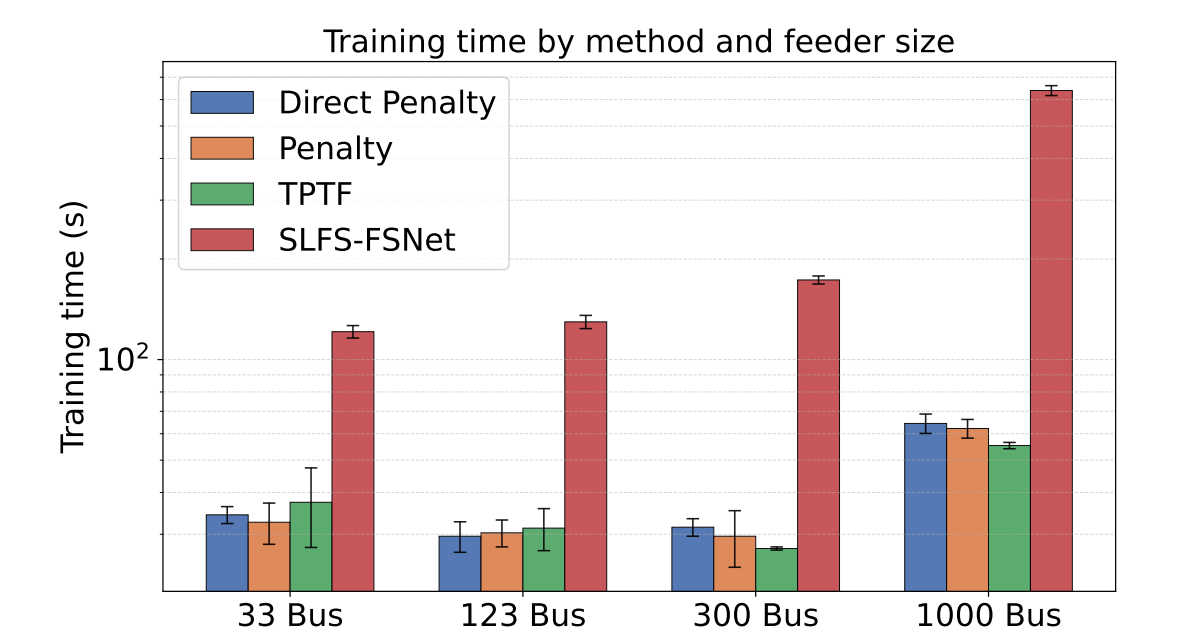


Table 1. Test results on AC-OPF with different methods over 5 random seeds

Method	Mean Violation	Max Violation	Optimality Gap(%)	Speedup
33 bus				
Direct Penalty	5.41e-03	0.026	-202.387	75.333
Penalty	7.51e-07	1.06e-03	0.024	37.049
TPTF	2.07e-08	4.16e-05	0.026	10.787
SLFS-FSNet	2.47e-07	4.86e-04	0.059	2.666
IPOPT	1.07e-08	5.96e-08	—	—
123 bus				
Direct Penalty	0.016	0.062	-105.792	149.328
Penalty	1.68e-05	2.77e-03	0.014	68.765
TPTF	1.07e-08	2.03e-05	0.087	16.350
SLFS-FSNet	8.63e-09	2.07e-05	0.103	16.054
IPOPT	1.30e-08	1.19e-07	—	—
300 bus				
Direct Penalty	0.010	0.066	-72.833	1004.152
Penalty	2.77e-06	9.98e-04	0.024	444.355
TPTF	2.61e-08	9.20e-05	0.026	119.506
SLFS-FSNet	1.11e-08	3.57e-05	0.084	116.305
IPOPT	1.18e-08	1.19e-07	—	—
1000 bus				
Direct Penalty	0.089	0.164	-13.274	3820.750
Penalty	1.26e-05	2.56e-03	3.563	1771.167
TPTF	2.51e-07	1.74e-04	3.510	346.377
SLFS-FSNet	7.69e-08	2.00e-04	0.402	384.482
IPOPT	1.15e-08	1.19e-07	—	—

Ongoing Work

- Consider generalization across changing topologies
- Extend the framework to multi-phase unbalanced systems with multi-period constraints